

Effectiveness of a tele-pulmonary rehabilitation program in individuals at risk of chronic obstructive pulmonary disease from artificial intelligence predictions

Busaba Chuatrakoon^{1*}, Siriphun Kongsawadsi¹, Kittichai Wantanajittikul², Sompong Sriburee², Supatcha Konghakote¹

¹Integrated Neuro-Musculoskeletal, Chronic Disease, and Aging Research Engagement Center (ICARE Center), Department of Physical Therapy, Faculty of Associated Medical Sciences, Chiang Mai University, Thailand.

²Department of Radiology, Faculty of Associated Medical Sciences, Chiang Mai University, Chiang Mai, Thailand.

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ABSTRACT

Background: Chronic Obstructive Pulmonary Disease (COPD) is a major global health burden, and early intervention in at-risk individuals may improve outcomes. Emerging technologies such as artificial intelligence (AI) offer novel means of identifying individuals at risk of COPD, enabling targeted preventive strategies. Tele-pulmonary rehabilitation (Tele-PR) has shown promise in managing chronic respiratory conditions, but its effectiveness in AI-identified at-risk individuals remains underexplored.

Objectives: To evaluate the effectiveness of a 12-week Tele-PR program in improving disease-related symptoms, muscular strength, and exercise capacity in individuals identified as at risk of COPD by AI prediction.

Materials and methods: Out of 286 screened participants, 38 were identified by AI as being at risk of COPD and enrolled in the study. They were allocated by village-level cluster assignment (quasi-experimental design) to either an intervention group (N=20) or a control group (N=18). The intervention group participated in a Tele-PR program conducted three sessions per week for 12 weeks. The control group received only verbal information on COPD and preventive self-management. Outcome measures included the modified Medical Research Council (mMRC) dyspnea scale, COPD Assessment Test (CAT), back and leg muscle strength using a dynamometer, and six-minute walk distance (6MWD). Assessments were conducted at baseline and post-intervention.

Results: All participants completed the study without adverse events. Following the intervention, the Tele-PR group showed a statistically significant improvement in 6MWD compared to the control group ($p=0.011$). However, no significant differences were observed between groups in mMRC scores, CAT scores, or muscle strength measurements.

Conclusion: A 12-week Tele-PR program significantly enhances exercise capacity, as measured by 6MWD, in individuals identified by AI as being at risk for COPD. These findings support the use of Tele-PR as an early intervention strategy in high-risk populations.

* Corresponding contributor.

Author's Address: Integrated Neuro-Musculoskeletal, Chronic Disease, and Aging Research Engagement Center (ICARE Center), Department of Physical Therapy, Faculty of Associated Medical Sciences, Chiang Mai University, Thailand.
E-mail address: busaba.c@cmu.ac.th
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Introduction

Chronic obstructive pulmonary disease (COPD) is a significant public health problem and the fourth leading cause of death worldwide.¹ It is characterized by persistent respiratory symptoms and airflow limitations that are progressive and not fully reversible.² Not only impaired respiratory function, but also muscle dysfunction, balance impairment, decreased exercise capacity, and

poor quality of life (QoL) are well recognized in individuals with COPD.^{3,4} Diagnosis of COPD is based on respiratory symptoms and confirmation of airflow limitation using spirometry. An assessment to diagnose COPD involves consideration of personal history and risk factors, along with additional investigations such as chest radiographs, CT scans, and arterial blood gas analyses.²

During the COVID-19 pandemic, spirometry was often restricted due to infection control concerns. Consequently, alternative screening approaches have gained attention. Recently, intelligent systems applying machine learning techniques to chest radiographs have been developed to estimate the likelihood of obstructive pulmonary disease. Such systems may facilitate early identification of individuals at increased risk of COPD in community settings, particularly where access to spirometry is limited.

COPD imposes a substantial burden on patients, families, and healthcare systems due to its chronic and progressive nature. Most previous studies have focused on rehabilitation strategies in patients with clinically established COPD.⁵⁻⁸ In contrast, little attention has been given to preventive interventions targeting individuals identified as being at increased predicted risk before clinically overt airflow limitation develops.

Pulmonary rehabilitation (PR) is strongly recommended by the Global Initiative for Chronic Obstructive Lung Disease (GOLD) as an essential component of COPD management.² Evidence demonstrates that PR improves exercise capacity, physical fitness, and health-related quality of life.^{6,7} However, access to center-based PR programs may be limited. Tele-pulmonary rehabilitation has emerged as an alternative approach and has been shown to provide benefits comparable to hospital-based programs in patients with COPD.⁸

Unlike previous studies evaluating telerehabilitation in confirmed COPD populations, the present study integrates an AI-assisted radiographic screening system with a preventive tele-pulmonary rehabilitation intervention in community-based adults identified as being at increased predicted risk of COPD. This study centers on two key components: (1) AI-assisted risk stratification to identify individuals prior to clinically established disease, and (2) implementation of a structured tele-pulmonary rehabilitation program in this at-risk population. By combining early risk identification with remote functional training, this approach shifts the focus from disease management to early functional preservation.

Therefore, this study aimed to examine the effects of a 12-week tele-pulmonary rehabilitation program (Tele-PR) on disease-related symptoms, muscle strength, and exercise capacity in individuals at increased predicted risk of COPD.

Materials and methods

Participants

A total of 286 community-based adults who had previously undergone chest radiography as part of

their annual health check-up were screened using the AI-based model. Among them, 38 individuals with predicted COPD risk scores ≥ 0.75 were identified. These individuals were subsequently assessed against the predefined inclusion and exclusion criteria. All 38 participants met the eligibility criteria and were enrolled in the study. The inclusion criteria were a minimum age of 40, the ability to comprehend the training program, and the capacity to communicate well with telemonitoring. Exclusion criteria were individuals with a history of home mechanical ventilator use, neurological and musculoskeletal conditions affecting postural control, symptomatic cardiovascular diseases, uncorrected visual problems, and non-smartphone users. The study protocol was approved by the Human Ethical Review Board of the primary investigator's institution (COAAMSEC-65EX-019). All eligible participants signed an informed consent form prior to enrollment in the study.

Study design

This study was a community-based quasi-experimental controlled study with cluster allocation at the village level. Four villages were included; two villages were assigned to the Tele-PR intervention group and two villages to the control group to minimize contamination between participants. No individual randomization was performed.

Sample size calculation

The sample size calculation in this study was based on a previous study that investigated the effect of tele-pulmonary rehabilitation in patients with COPD on their exercise capacity as measured by a 6-minute walk test (6MWT).⁹ The mean and standard deviation of 6MWT between the intervention and control groups were 420.2 ± 74.9 and 339.9 ± 110.1 , respectively. This yielded an effect size of 0.85. With 80% power and a 5% type-I error, a minimum sample size of 36 participants was required to detect a between-group difference in 6MWD. Considering a possible dropout rate of 10%, 40 participants (20 per group) were enrolled in the study.

Instrument

AI-assisted screening tool for individuals at risk of COPD

A deep learning-based screening model was developed using a transfer learning approach with a pretrained ResNet-50 convolutional neural network architecture.¹⁰ The model was implemented in MATLAB (Version 2023b, The MathWorks, Inc., Natick, MA, USA).

For model development, 540 labeled digital chest radiographs from normal individuals and patients across different COPD severity stages were used for supervised training, with an additional 60 radiographs used for internal validation. These datasets were independent of the participants included in the present intervention study. The trained model generated a probability score representing the likelihood of radiographic patterns

associated with airflow limitation. Feature extraction was performed automatically through convolutional layers without predefined radiographic variable selection.

In the present study, a separate cohort of 286 community-based adults who had undergone annual chest radiography was screened using the trained model. From this independent screening cohort, 38 individuals with probability scores ≥ 0.75 were enrolled. The probability score was applied as a risk stratification tool to identify individuals with relatively high predicted

risk. A relatively high probability threshold (0.75) was selected to reduce potential false-positive classification and to focus on individuals with more pronounced radiographic patterns suggestive of increased risk. This threshold was pragmatically defined for inclusion in this exploratory preventive study and does not represent a clinically validated diagnostic cut-off. External validation in independent populations is warranted to further establish the model's generalizability (Figure 1).

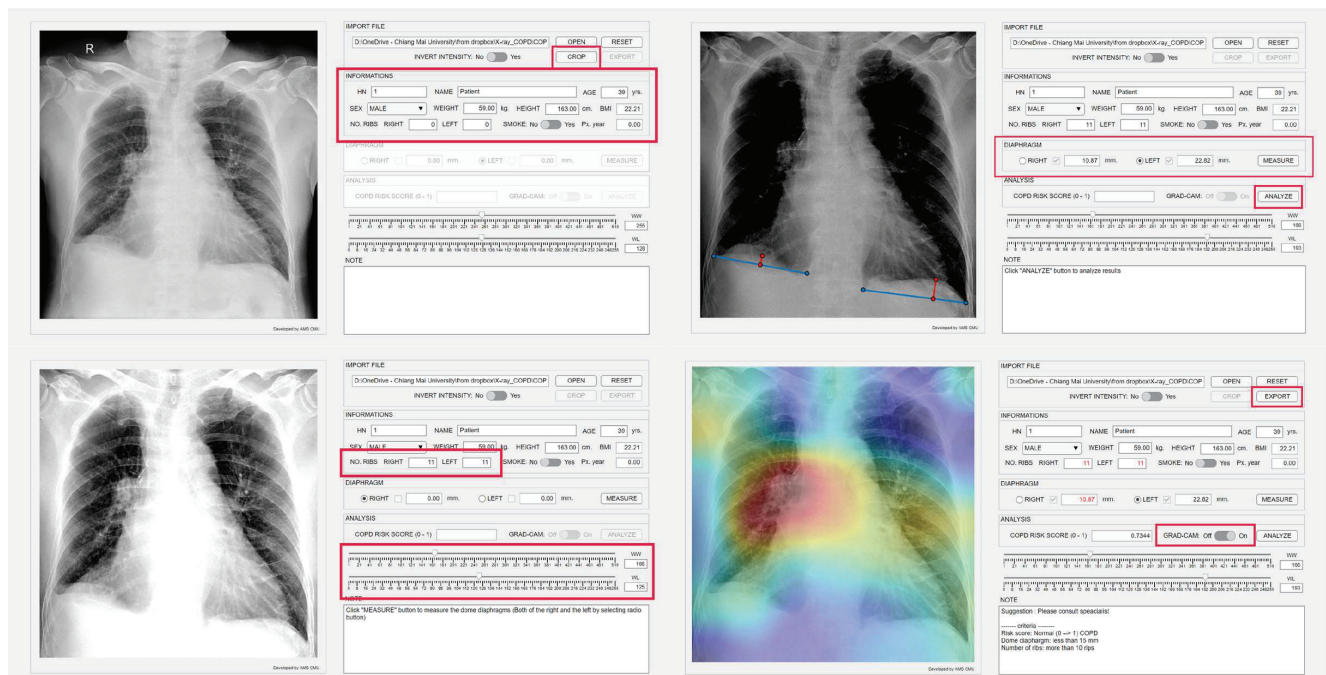


Figure 1. A deep learning-based screening system for the risk of COPD.

Tele-pulmonary rehabilitation (Tele-PR) program

In this study, a home-based Tele-PR program was delivered via the Line application on a smartphone platform. Participants in the intervention group performed the program three times per week for 12 weeks. Each session lasted approximately 45-60 minutes and consisted of 1) breathing and chest mobilization exercises (approximately 10 minutes) to enhance ventilatory mechanics, 2) stretching exercises (approximately 10 minutes) to promote flexibility, 3) resistance exercises for upper and lower limbs using elastic bands (15-20 minutes) to improve muscle strength, and 4) aerobic walking training (15-20 minutes) to enhance cardiopulmonary endurance. Each session consisted of structured exercise components,

supplemented by self-management education and adherence support delivered through instructional materials and weekly telemonitoring via the Line application.

At the initial visit, participants received face-to-face instruction on proper exercise techniques and application use. Written and video instructional materials were provided through the Line application. The program consisted of three progression levels, which were advanced every four weeks according to participant tolerance. Exercise intensity was monitored using the Borg Rating of Perceived Exertion scale, targeting levels 3-5 for strengthening exercises and 3-6 for aerobic training (Table 1).¹¹

Table 1. Details of the tele-pulmonary rehabilitation program (12 weeks, 3 sessions/week).

Intervention components	Level 1 (Weeks 1-4)	Level 2 (Weeks 5-8)	Level 3 (Weeks 9-12)
Breathing exercise	6 breaths/set, 5 sets/session	Same as Level 1	Same as Level 1
Stretching exercise (8 muscle groups)	10 second hold × 5 repetitions	Same as Level 1	Same as Level 1
Strengthening exercise (6 muscle groups, elastic band, Borg 3-5)	10 reps × 2 sets	10 reps × 3 sets (increased resistance)	12 reps × 3 sets (increased resistance)
Endurance exercise (walking)	Borg 3-4, 15 minutes	Borg 4-5, 20 minutes	Borg 5-6, 25 minutes
Self-management and education	Initial instruction + weekly reinforcement	Ongoing reinforcement	Ongoing reinforcement
Estimated session duration	~45-60 minutes	~45-60 minutes	~45-60 minutes

Note: reps: repetitions.

Resistance was progressed by selecting elastic bands with greater tension when participants were able to complete the prescribed repetitions within the target Borg RPE range (3-5) without excessive fatigue. If participants reported RPE below the target range, resistance was increased in the subsequent session; if RPE exceeded the target range, resistance was adjusted downward. Stretching exercises were performed with a 10-second hold per repetition, repeated five times per

muscle group.

Weekly follow-up messages were sent to reinforce adherence, and participants were required to report exercise completion through the application. Participants were instructed to discontinue exercise and contact the research team if they experienced chest pain, dizziness, severe dyspnea, or other abnormal symptoms. No adverse events were reported during the intervention period (Figure 2).

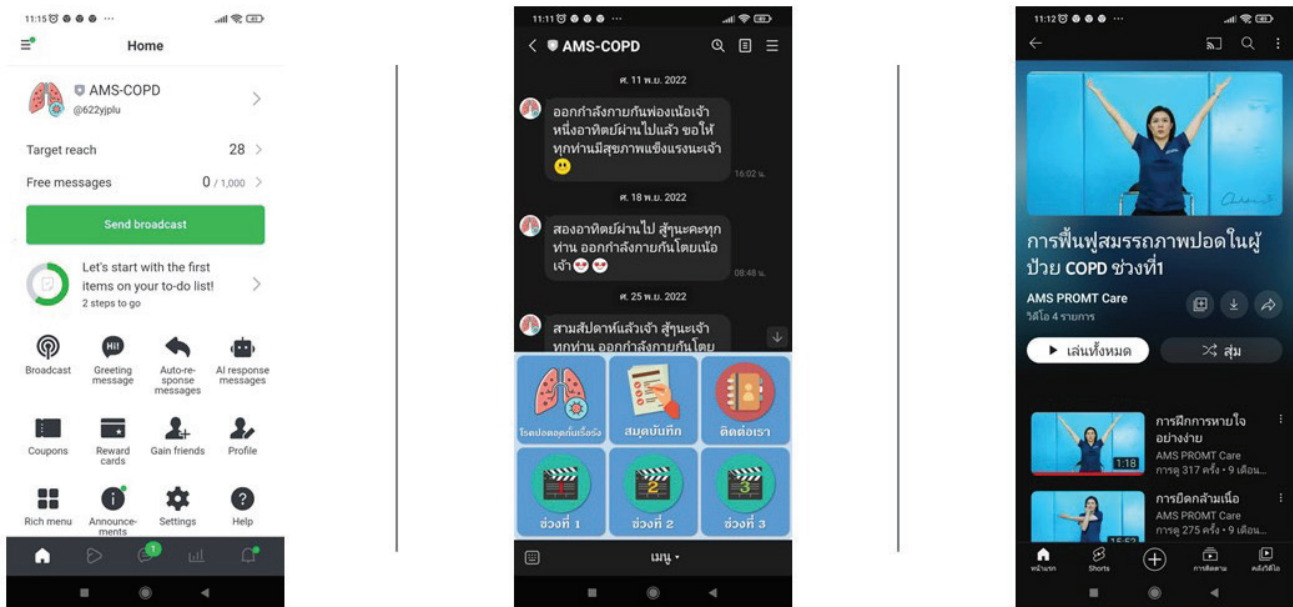


Figure 2. The Tele-PR via Line application.

Baseline characteristics and outcome measures

Participant characteristics, spirometry, and all outcome measures were assessed using standard, structured protocols at baseline and post-12-week intervention by trained assessors. Airflow obstruction was defined as $FEV_1/FVC < 0.70$.

Outcome measures

The primary outcome of this study was exercise capacity measured by the Six-Minute Walk Distance (6MWD).

The Six-Minute Walk Test (6MWT) was conducted according to standard guidelines.¹² Participants walked at their own pace along a 30-meter corridor for six minutes, and the total walking distance (meters) was recorded for analysis. The 6MWD was selected as the primary outcome due to its established validity and clinical relevance in evaluating functional exercise capacity in respiratory populations.

Secondary outcomes included symptom burden, health-related quality of life, and muscle strength:

The COPD assessment test-Thai version (CAT-TH), a short, simple, valid, and reliable self-administered questionnaire, was used to determine health-related quality of life. High CAT scores indicate low QoL.¹³

The modified Medical Research Council dyspnea scale (mMRC) was used to identify disability levels related to dyspnea. It is a five-point Likert scale (0-4), with a higher score indicating greater breathlessness. The mMRC has good test-retest reliability.¹⁴

A back-leg dynamometer was used to measure the back and leg muscles' strength. The resisted force was recorded in kilograms.¹⁵

Study procedure

Participants were recruited through verbal announcements and poster advertisements within the

community. After providing written informed consent, their existing chest radiographs were analyzed using the trained AI-based screening model. Individuals with AI-predicted COPD risk scores ≥ 0.75 were contacted by telephone to confirm eligibility.

Thirty-eight eligible participants were scheduled for baseline assessment at the community health center prior to group allocation. The intervention group (N=20) received the Tele-PR program delivered through the Line application with remote monitoring for 12 weeks. The control group (N=18) received general primary healthcare education delivered by community healthcare personnel. The educational content included advice on smoking cessation, physical activity promotion, respiratory hygiene, and chronic disease prevention. Education was provided in a standard informational format without individualized exercise prescription or supervised training. Participants in the control group were advised to continue their usual daily activities during the study period. All participants were assessed on outcome measures at baseline and after the 12-week intervention period.

Data analysis

Normality of data distribution was assessed using the Shapiro-Wilk test. Baseline demographic characteristics between groups were compared using independent t-tests for continuous variables and chi-square tests for categorical variables. Between-group differences in changes from baseline were analyzed using independent t-tests. Continuous variables are presented as means with 95% confidence intervals. A two-tailed significance level of $p < 0.05$ was considered statistically significant. Data were analyzed using SPSS software (version 21.0, IBM Corporation, Chicago, IL, USA).

Results

Participants' characteristics

Of the 286 participants in the communities, 38 were eligible and allocated either to intervention or control groups. No differences in demographic characteristics between the two groups were observed at baseline (Table 2). Baseline pulmonary function, characterized by FEV₁/FVC ratio and % predicted FEV₁, did not differ

significantly between groups. Raw FVC values were not available from the archived dataset. The term “spirometry-defined airflow obstruction” refers to participants with FEV₁/FVC <0.70 and does not indicate a clinical diagnosis of COPD. All participants completed the training program with no reported adverse events. The average exercise attendance rate was 92%.

Table 2. Demographic characteristics of the participants (N=38)

Variables	Tele-PR (N=20)	Control (N=18)	p value
Gender (male, %)	7 (35)	4 (22.2)	0.485
Age (year)	58.3±9.5	52.7±8.1	0.059
AI-risk score (%)	0.9±0.1	0.9±0.1	0.644
Height (m)	1.6±0.1	1.6±0.1	0.533
Weight (kg)	55.8±8.9	54.9±8.4	0.762
BMI (kg/m ²)	22.7±3.3	21.8±2.2	0.336
Smoking (N, %)	3 (15)	2 (11.1)	1.000
Occupational pollution time (hour)	2.1±2.0	1.4±1.3	0.202
FEV ₁ /FVC	78.8±7.0	83.5±7.7	0.054
Predicted FEV ₁ (%)	88.5±10.9	90.0±16.4	0.740
Spirometry-defined airflow obstruction (FEV ₁ /FVC<0.70, N (%))	3 (15)	2(11.1)	1.000

Note: AI: artificial intelligence, BMI: body mass index, FEV₁: forced expiratory volume in the first second, Tele-PR: tele-pulmonary rehabilitation program.

Effectiveness of the Tele-PR program

Descriptive data of all outcome variables from baseline to the post-intervention period are presented in Table 3. Baseline comparisons of all outcome variables between groups revealed no statistically significant differences (all $p>0.05$). Baseline back and leg strength

values appeared to be within expected ranges for community-based adults of similar age, suggesting relatively preserved musculoskeletal function. A significant improvement in 6MWD was observed in the intervention group compared to the control group after the 12-week training period ($p=0.011$; effect size =0.87).

Table 3. Outcome measures for the intervention and control groups.

Variables	Tele-PR (N=20)			Control (N=18)			95% Confidence Interval	p value	Effect size
	Pretest	Posttest	Change	Pretest	Posttest	Change			
CAT score (point)	4.1±4.1	2.2±4.9	-1.9±6.6	3.3±3.3	2.8±3.5	-0.6±2.5	-2.01 to 4.70	0.422	0.26
mMRC (level)	0(0.75)	0(0)	0(0)	0(0.25)	0(0)	0(0)	-0.35 to 0.49	0.747	-
Back strength (kg)	65.5±28.8	70.8±31.8	5.3±11.0	54.9±22.7	59.1±20.1	4.2±5.2	-6.91 to 4.65	0.693	0.13
Leg strength (kg)	72.8±34.3	75.6±34.0	2.9±11.1	67.8±29.0	71.4±22.8	3.6±11.9	-6.86 to 8.28	0.850	0.06
6MWD (m)	428.5±57.7	466.9±70.3	38.4±40.9	430.9±34.1	441.2±37.7	10.3±19.8	-49.16 to -7.08	0.011*	0.87

Notes: * $p<0.05$ difference between the change score of tele-PR and control groups. CAT: COPD assessment test, mMRC: modified medical research council dyspnea scale, Tele-PR: tele-pulmonary rehabilitation, 6MWD: 6-minute walk distance.

Discussion

Pulmonary rehabilitation (PR) has been widely recommended as an effective strategy for individuals with COPD. In addition to center-based programs, growing evidence supports the effectiveness of home-based and tele-pulmonary rehabilitation interventions.^{5,16-18} However, most previous studies have focused on patients with established COPD across varying severity stages.^{5,6,16-18} The present study differs conceptually in that it targeted community-based adults identified as having an increased predicted risk of COPD using an AI-assisted radiographic screening system. Rather than focusing on established diseases, this study explored a preventive approach aimed at promoting functional capacity before substantial physiological decline occurs. Although baseline spirometry parameters and symptom scores were generally within normal ranges, indicating that most participants did not have clinically established COPD, the AI-derived risk score may capture subtle structural radiographic changes that precede clinically detectable airflow limitation. Therefore, the intervention targeted individuals with increased predicted risk rather than confirmed COPD, emphasizing early health promotion.

This study is novel in that it integrates AI-assisted radiographic risk stratification with a structured tele-pulmonary rehabilitation program in a preventive community-based setting. Unlike previous studies that focused on patients with established COPD, this approach applies artificial intelligence to identify individuals at increased predicted risk and implements early functional intervention before clinically overt disease develops. To our knowledge, this combined screening-intervention framework has not been previously reported in community-based populations.

The main finding demonstrated that the 12-week Tele-PR program significantly improved exercise capacity compared to education alone. The mean improvement in 6MWD (38.4 meters) was not only statistically significant with a large effect size but also exceeded commonly reported minimal clinically important difference (MCID) thresholds of approximately 25-35 meters in respiratory populations.¹⁹ Although MCID values have primarily been established in patients with chronic respiratory disease, exceeding these thresholds suggests that the improvement is likely to be perceptible and functionally meaningful in daily activities, even in individuals without established COPD. In a preventive population, such enhancement in walking capacity may reflect improved physical resilience and reduced risk of future functional decline. The improvement in exercise capacity may be attributed to enhanced cardiovascular endurance, peripheral conditioning, and improved exercise adherence supported by structured telemonitoring.

Given the relatively preserved baseline lung function and low symptom burden, the observed functional gains likely reflect improved submaximal exercise efficiency rather than measurable changes in pulmonary mechanics. This may explain why significant

changes were observed only in exercise capacity, whereas symptom scores and muscle strength demonstrated limited improvement. Exercise intensity during training was monitored using the Borg scale; however, RPE and SpO₂ were not systematically recorded during the 6MWT, which limits further physiological interpretation.

In contrast, disease-related symptoms, muscle strength, and health-related quality of life showed only slight improvements in both groups without significant differences. Baseline CAT and mMRC scores were low, indicating minimal symptom burden at study entry. Although these measures were included to explore potential subclinical symptom changes in individuals at increased predicted risk of COPD, the limited improvement observed is consistent with the preventive stage of this population. A potential ceiling effect may therefore explain these findings, as participants demonstrated relatively preserved baseline lung function and low symptom severity. Baseline back and leg strength values appeared to be within expected ranges for community-based adults of similar age, suggesting relatively preserved musculoskeletal function at baseline. Furthermore, the strengthening component of the Tele-PR program was adapted from protocols originally designed for symptomatic COPD populations and may not have provided sufficient overload stimulus for this relatively healthy group.

From a public health perspective, the findings suggest that tele-pulmonary rehabilitation may serve not only as a therapeutic strategy for established COPD but also as a preventive health promotion tool in individuals at increased predicted risk. Early functional enhancement may potentially contribute to long-term reductions in healthcare utilization and associated costs, although longitudinal studies are required to confirm this hypothesis. While improvements in 6MWD may be expected with regular exercise, pulmonary rehabilitation is a structured, progression-based intervention that integrates respiratory techniques, aerobic conditioning, resistance training, and adherence support. In individuals at increased predicted risk of COPD, early implementation of such a structured program may enhance functional reserve and promote long-term physical resilience before clinically significant physiological decline occurs. This preventive framework distinguishes the present study from therapeutic rehabilitation in established COPD populations. Regarding the AI-assisted screening model, it should be emphasized that the model was developed and internally validated during the training phase and was applied in this study as a risk stratification tool rather than a diagnostic instrument. Although promising as a screening support system, external validation in independent cohorts is required before broader clinical implementation.

Limitations

This study has several limitations. Firstly, the absence of a follow-up period limits conclusions regarding the long-term sustainability of the intervention

effects. Secondly, spirometry was not reassessed post-intervention, as the primary focus was functional exercise capacity rather than short-term pulmonary function change. Thirdly, baseline habitual physical activity levels were not objectively assessed, and differences in daily activity patterns between groups cannot be completely excluded. Although baseline functional capacity and muscle strength did not differ significantly, unmeasured variability in habitual activity may have influenced the magnitude of response to the intervention. Lastly, the strengthening intensity may not have been sufficient for participants with relatively preserved baseline musculoskeletal function. Future studies should consider individualized progression strategies, objective monitoring of physical activity levels, longitudinal follow-up, and external validation of the AI-based screening model.

Conclusion

The 12-week tele-pulmonary rehabilitation program significantly improved exercise capacity in community-based adults identified as having increased predicted risk of COPD through an AI-assisted radiographic screening approach. Although participants did not have established COPD, the findings suggest that early, structured tele-rehabilitation may enhance functional capacity in individuals with elevated predicted risk. This preventive strategy may contribute to promoting long-term respiratory health in community settings. Further research is warranted to optimize intervention intensity, evaluate long-term outcomes, and externally validate the AI-based risk stratification model.

Ethical approval

The study was performed in compliance with relevant laws and institutional guidelines, and the appropriate institutional committees approved it (COAAMSEC-65EX-019); participants also signed a written consent form.

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This research received no external funding.

Conflict of interest

The authors report there are no conflicts of interest in this work.

Data availability statement

The data presented in this study are available upon request from the corresponding authors.

CRedit authorship contribution statement

Busaba Chuatrakoon: conceptualization, methodology, investigation, data collection, data curation, writing original draft, review and edit, supervision; **Siriphun Kongsawasdi:** methodology, data curation, review and edit; **Kittichai Wantanajittikul:** data curation, review and edit; **Sompong Sriburee:** data collection, review

and edit; **Supatcha Konghakote:** data collection, review and edit.

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